

Machine Learning 1

Assessment rules

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academic year 2025/2026

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General information

In teams of **at most 2 persons students** will work on two practical machine learning projects - one for **regression** and one for **classification**. Please inform the lecturer about the team members by email pwojcik@wne.uw.edu.pl or sm.lis@student.uw.edu.pl **the latest by midnight 2026-04-20**.

Datasets

The data is **exactly the same for all teams**.

restaurants for **classification**

Your task is to apply various ML algorithms and tools **discussed on Machine Learning 1 course** (see the rules below) to build a model **explaining the restaurant survival**

based on the **training sample** and generate predictions for **all observations** from the **test sample**.

More details are provided here: <https://www.kaggle.com/competitions/ml-1-2026-task-1-predicting-restaurant-survival-classification>

Files Provided

- `restaurants_train.csv` - training data contains 33296 observations and 86 columns along with the target variable **status_closed**.
- `restaurants_test.csv` - test data contains 8325 observations and 85 columns **without** the target variable.
- `restaurant_sample_submission.csv` - sample submission in the correct format file containing 8325 observations and 2 columns: `restaurant_id` and `status_closed` (prediction).

`annual.pay.usd` for regression

Your task is to apply various ML algorithms and tools discussed on Machine Learning 1 course (see the detailed requirements below) to build a model **predicting the annual compensation of software developers** based on the **training sample** and generate predictions for **all observations** from the **test sample**.

More details are provided here: <https://www.kaggle.com/competitions/ml-1-2026-task-2-developer-salary-prediction-regression>

Files Provided

- `train.csv` – training data contains 2,512 observations and 41 columns along with the target variable **annual.pay.usd**.
- `test.csv` – test data contains 628 observations and 41 columns (40 features + id) **without** the target variable.
- `sample_submission.csv` – sample submission file in the correct format.

Requirements

1. **Exploratory Data Analysis:**
 - Analyze the dataset to identify key patterns, correlations, and potential challenges introduced by the additional variables.
 - Visualize distributions and relationships among variables.
2. **Feature Engineering:**
 - Consider transformation or scaling of variables as needed.
 - Evaluate the impact of the supplementary variables on model performance.
3. **Modeling:**
 - Build, train, and compare multiple regression models.
 - Optimize model hyperparameters using cross-validation.
4. **Predictions:**

- Generate predictions for all observations in the test dataset.
 - Document model performance and reasoning behind the selected approach.
5. **Documentation:**
- Provide a clear explanation of your analysis, modeling choices, and any challenges faced while integrating the additional variables.

Competitions on kaggle

Both datasets and their descriptions are available on kaggle:

- restaurants for classification: <https://www.kaggle.com/competitions/ml-1-2026-task-1-predicting-restaurant-survival-classification>
- annual.pay.usd for regression: <https://www.kaggle.com/competitions/ml-1-2026-task-2-developer-salary-prediction-regression>

In both cases you are required to upload the file with final predictions that **only includes** the observation **id** and the **predicted value of the outcome variable** (check details on kaggle).

IMPORTANT! Groups that do NOT upload their submissions on time **will not be graded**.

- in case of each dataset **30% of the test data** will be used for the **Public Leaderboard** during the competition,
- the remaining **70% of the test data** will be used for the **final Private Leaderboard** - used in the **course assessment**

Various algorithms

For each of the datasets please consider and compare **at least 3 different ML algorithms discussed in the ML1 course** (e.g. linear/logistic regression, KNN, LASSO, ridge, elastic net, SVM/SVR with various kernel functions). Do **NOT** use any other algorithms beyond the scope of the ML1 course. Still you can apply other known tools - data rebalancing, feature engineering, feature selection.

Selection of the best algorithm

The choice of the final algorithm applied to generate predictions should be **clearly explained** in the presentation.

HINT !!!!! Use the internal division of the training data into train/validation/test samples to make sure that you correctly assess the performance of your models on the new data.

Performance measure

The performance of predictions will be based on:

1. balanced accuracy for the **restaurants** dataset
2. RMSE for the **annual.pay.usd** dataset

Please report the **expected** value of a particular performance measure (expectation for the test sample) in your presentation. It can be based on the public leaderboard from kaggle.

Points

In total **60 points** can be collected - 30 for each project:

- **presentation in class** - its structure, way of presenting, etc. (**10 pts** for each project)
- **presentation contents** - assessed by the lecturer after you present in class (**10 pts** for each project)
- **out-of-sample** performance (**10 pts** for each project):
 - 10 if predictive performance in top quartile group (best),
 - 7.5 if predictive performance in the 2nd quartile group (good),
 - 5 if predictive performance in the 3rd quartile group (below average),
 - 2.5 if predictive performance in the 4th quartile group (unlucky),

Presentations

The competitions on kaggle are open **until 23:55 2026-05-28**. By this deadline you are **required** to **upload your team's submission** file on kaggle. The same deadline applies to the submission if the **presentation** together with **the full codes**. Please do it by email to the lecturer pwojcik@wne.uw.edu.pl or sm.lis@student.uw.edu.pl or share your Github account. The codes should **load the training data**, apply data transformation/selection, **train** the single best algorithm, **apply** this model on the **test data** and save test data predictions in the `csv` file.

Separately share the codes which you applied to find the best algorithm, parameter search, etc.

All teams will give presentations (not more than **10 minutes**) informing about the algorithms considered, selection process and their **expected results**.

Presentations will take place on 2026-05-29 for groups taught by Szymon Lis and Michał Woźniak and on 2026-06-01 for groups taught by Piotr Wójcik.

Groups that do NOT present their results in class **will not be graded**.

Important dates again

- 2026-04-20 by end of day - submission of information about the **team members**
- 2026-05-28 by 23:55 - submission of **presentations**, **codes** and uploading **test sample predictions** on kaggle,
- 2026-05-29 or 2026-06-01 - in class **presentations** - depending on the group.

GOOD LUCK and happy modeling !!!!!!!!!!!!!!!